

DEVANSH KADU

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PROFESSIONAL SUMMARY

Computer Science Engineering (Data Science) student with expertise in **Object-Oriented Design, Operating Systems, Algorithms, and Data Structures**. Gained hands-on experience in **GPU-accelerated cryptographic systems** and encrypted database design through various projects, including a full-stack web application that enhanced real-time data management. Proven ability to apply complexity analysis in software development, improving workflows and system efficiency.

WORK EXPERIENCE

Software Developer Intern (SDE) | Internship Nov 2025 - Mar 2026
KraftX Works Hybrid • Pune

- **Architected full-stack web applications** using **React, TypeScript, Node.js**, and **Python**, enabling **real-time data management** with **Supabase** and **Firebase**.
- **Integrated Razorpay payment gateways and webhooks**, streamlining custom payment links and ensuring instant status synchronization and transaction workflows.
- **Engineered automated backend workflows** through **Google Cloud APIs**, reducing manual operations and enhancing system response times.
- **Collaborated with multiple clients** to gather functional requirements, manage project scope, and deliver production-ready web platforms on schedule.

Technologies / Skills Used : *React, Node.js, Python, TypeScript, JavaScript, Object-Oriented Design, Operating Systems, Algorithms, Data Structures, complexity analysis*

PROJECTS

Cedar - Identity and Access Management | 🐙 Github Dec 2025 - Mar 2026
Co-Developer

- Co-developed a **Rust CLI tool (cedar-policy, clap, anyhow)** in a **3-person team** to **streamline AWS Cedar policy debugging**.
- Automated combination testing across **2ⁿ policy permutations** using **game-theoretic algorithms** (Banzhaf metrics) and **attribute relevance scoring** to isolate decision-critical access controls.
- Applied **object-oriented design principles** (Rust traits/structs) to build an interactive mutation engine for **real-time policy edits**, request simulations, and rapid re-evaluation.
- Optimized execution to process **1,024 policy permutations (2¹⁰)** in **0.25 seconds** with a **14MB peak memory footprint**, surfacing explicit decision diagnostics for permit, forbid, and implicit-deny outcomes.

Technologies / Tools Used : *Rust, AWS Cedar (cedar-policy 2.5.1), serde_json, clap, anyhow*

WidViz – AI-Powered YouTube Video Summarizer & Study Companion | 🐙 Github Dec 2024 - Mar 2025
Co-Developer

- **Architected** WidViz, a locally hosted desktop application that consolidates fragmented study workflows into a single offline platform, **eliminating** reliance on cloud tools.
- **Developed** video processing pipelines to auto-summarize YouTube content and generate interactive quizzes, **enhancing** comprehension and content retention.
- **Engineered** real-time note-taking and daily goal-tracking modules to maintain self-contained study sessions within the platform.

Technologies / Tools Used : *Python, Flask, SQL, JavaScript, HTML, CSS*

AGNIYA – GPU-Accelerated Homomorphic Encryption Database | 🐙 Github Aug 2025 - May 2026
Co-Developer

- **Architected** a GPU-accelerated encrypted database using **Paillier-2048 homomorphic encryption**, achieving **6.7M cached ops/sec** and **2.7K ops/sec** on fully encrypted queries.
- **Engineered** CUDA GPU pipelines, **Numba JIT compilation**, and **AVX2 SIMD vectorization** to deliver **10-50x speedup** over standard Paillier operations, with automatic CPU fallback.
- **Designed** multi-level RAM/disk caching and **Path ORAM** access-pattern hiding, improving hot-data query latency by **up to 251x** while preserving semantic security.
- **Built** a full-stack platform using **Next.js, React, and Flask** for encrypted CSV ingestion, per-column encryption configuration, and encrypted SUM/AVG/COUNT querying.
- **Developed** RNS-based ciphertext compression and LSH-based encrypted deduplication, reducing bandwidth by **50%** and storage by **60-80%**, validated through a reproducible benchmark suite.

Technologies / Tools Used : *Python, CUDA, Numba, NumPy, gmpy2, Flask, Next.js, React, TypeScript*

SKILLS

Programming Languages :	C, C++, Python, JavaScript, TypeScript, SQL, Rust, HTML5, CSS3
Frameworks & Libraries :	React, Next.js, Node.js, Flask, FastAPI, clap, anyhow, cedar-policy, Tailwind CSS, Shadcn UI, NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Hugging Face, Numba
Computer Science Fundamentals :	Data Structures, Algorithms, Object-Oriented Design, Operating Systems, Complexity Analysis
Security & Cryptography :	Homomorphic Encryption, Paillier Cryptosystem, GPU-Accelerated Cryptography
Tools & Platforms :	Git, GitHub, Docker, Linux, CUDA, GPU Computing, gmpy2, Google Cloud APIs, Razorpay API & Webhooks, RESTful APIs
Databases :	PostgreSQL, MySQL, SQLite, MongoDB, Firebase, Supabase
Soft Skills :	Teamwork, Problem Solving, Critical Thinking, Time Management, Communication
Languages :	English, Hindi

EDUCATION

Vishwakarma Institute of Technology | Pune, India 2024 - 2028
B.Tech CSE(Data Science) CGPA : 8.59

Currently enrolled in the **B.Tech CSE (Data Science)** program at **VIT Pune (2024–2028)**, focused on building **production-grade full-stack applications** and **automated backend systems**.

My background spans the full development lifecycle—from client discovery and UI design in **React and TypeScript** to backend microservices in **Node.js and Python**. I've architected real-time application pipelines using **Supabase and Firebase**, integrated **Google Cloud APIs** for workflow automation, and engineered transaction engines using **Razorpay webhooks**. Working on client-facing applications taught me that the hardest part of software isn't stitching APIs together; it's **designing clean abstractions that survive real-world edge cases**.

Driven by a desire to understand why systems work (and how they break), I'm actively building software that **scales reliably from database to frontend**.